Musculoskeletal Engineering

Share Tech Mono

Operational Protocols for Crutches, Casts, & Joint Maintenance

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System Diagnostic Legend

Share Tech Mono
OPTIMAL STATE / STABLE

Share Tech Mono
CRITICAL FAILURE / DANGER

Share Tech Mono
CAUTION / WARNING

Share Tech Mono
MECHANIST INTERVENTION (ACTION)

Share Tech Mono
INTERNAL MECHANISM (PHYSIOLOGY)

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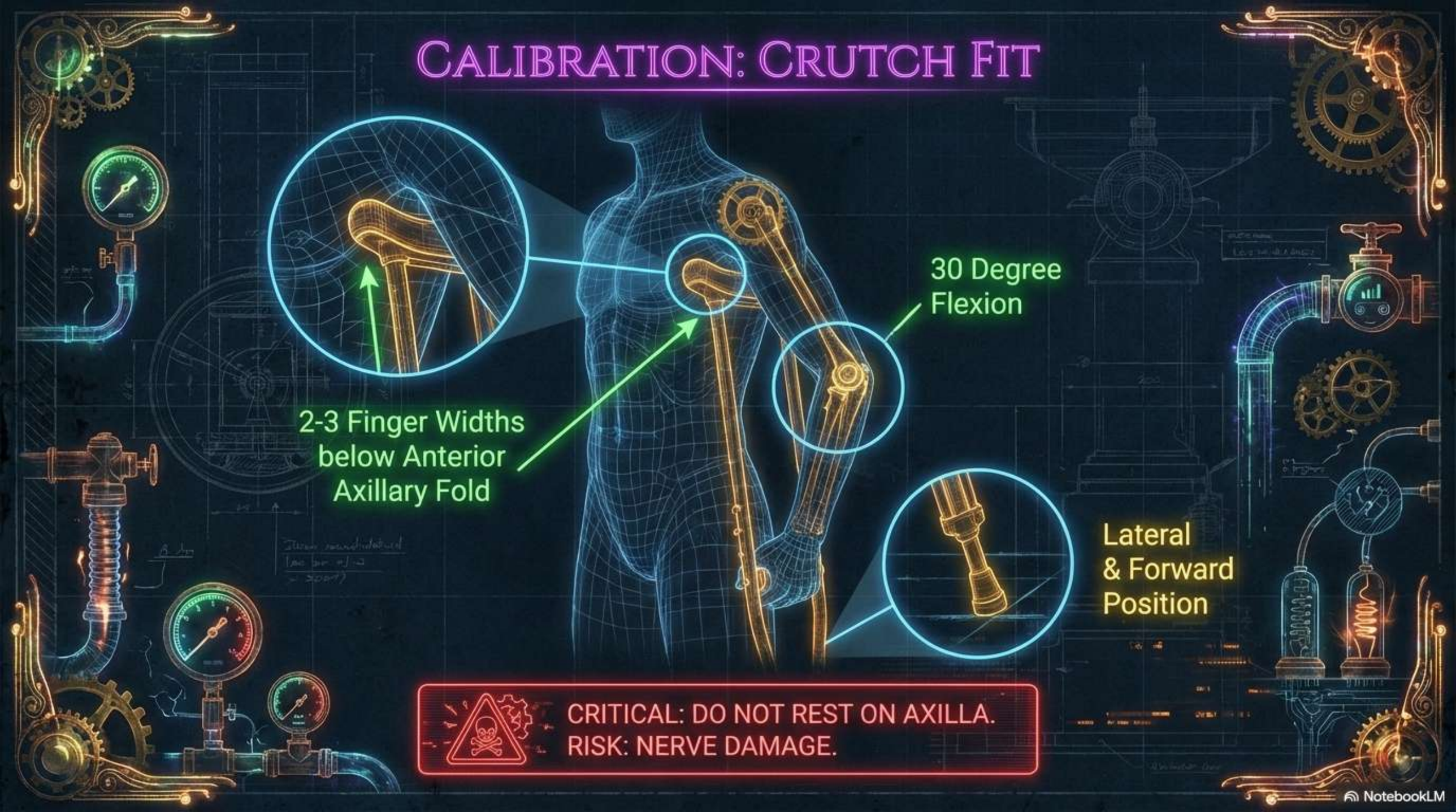
Calibration: Crutch Fit



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Locomotion Algorithm

EVEN for EVEN. ODD for ODD.

Input: Even Distribution
(Bilateral Weakness)

Input: Odd Distribution
(Unilateral/One Leg)

Output:
EVEN Gait
(2-Point or
4-Point)

Output:
ODD Gait
(3-Point)

System Note: Even numbers = even number of legs affected.

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Bilateral Weakness Protocols

2-POINT GAIT (MILD)

Move Crutch + Opposite Foot TOGETHER.

4-POINT GAIT (SEVERE)

Move Everything SEPARATELY.

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UNILATERAL & NON-WEIGHT BEARING

3-POINT GAIT

Move 2 Crutches + Bad Leg TOGETHER.

SWING-THROUGH

Target: Amputees / Non-Weight Bearing.

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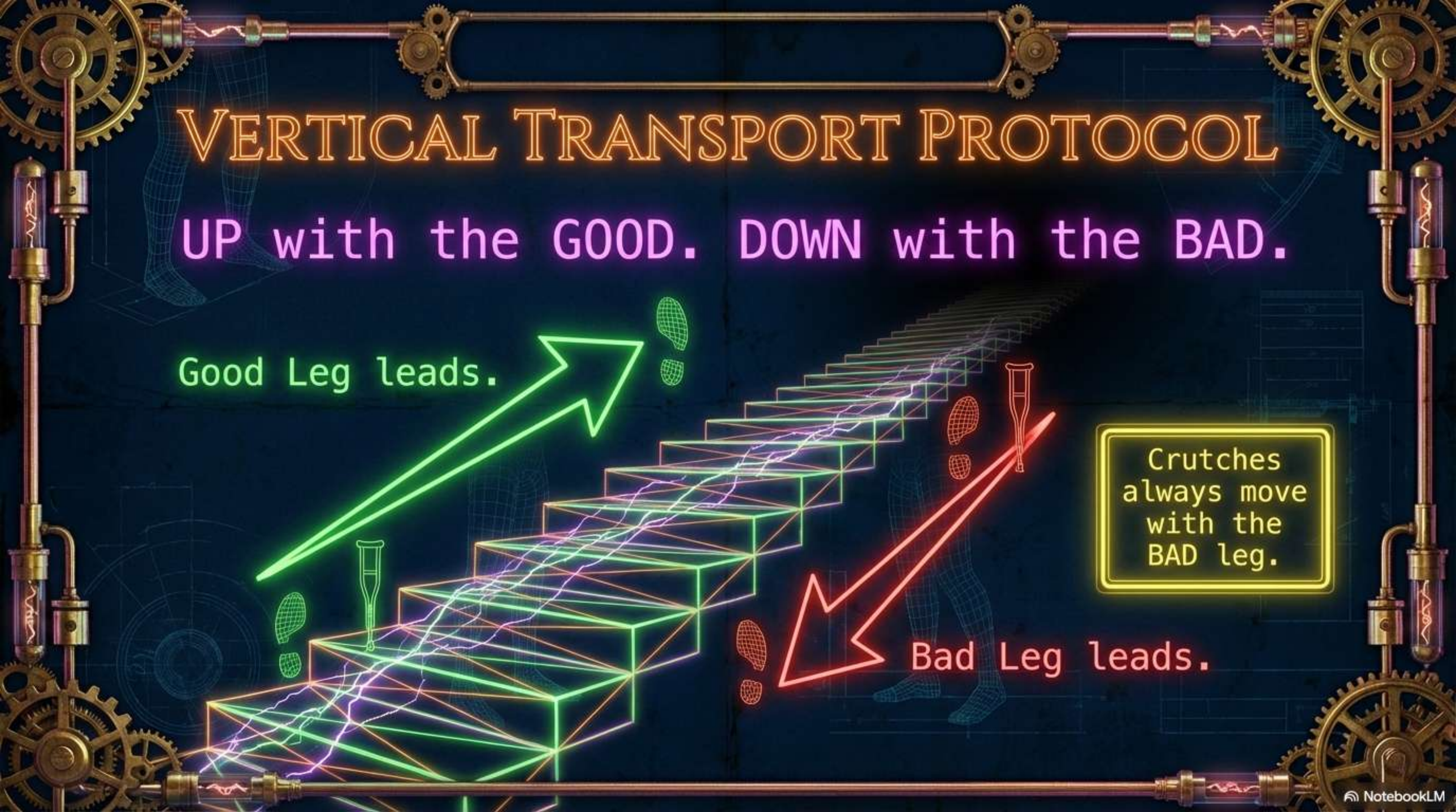
Vertical Transport Protocol

UP with the GOOD. DOWN with the BAD.



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AUXILIARY STABILIZERS

CANE

Hold on STRONG / UNAFFECTED Side.

Advance with Weak side.

WALKER

Pick it up, set it down, walk to it.

NO belongings tied to front (Tipping Hazard).

Stand up BEFORE grabbing walker.

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Structural Containment: Casting

Handling Wet Material:
PALMS ONLY.

No Fingertips (Prevents Indentations).

Curing Protocols

Air Dry / Fan OK.

No Heat Lamps / No Hair Dryers.

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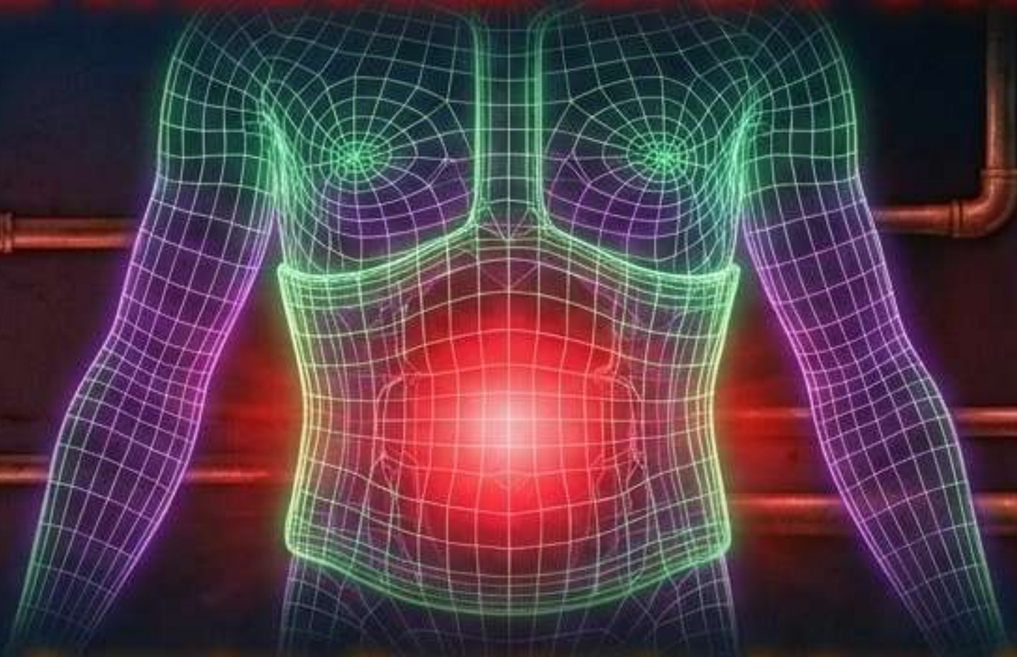
Critical Failure: Cast Syndrome



Mechanism: Superior Mesenteric Artery Compression.

Symptoms: Nausea, Vomiting, Distension.

INTERVENTION: 1. NPO 2. NG TUBE DECOMPRESSION.

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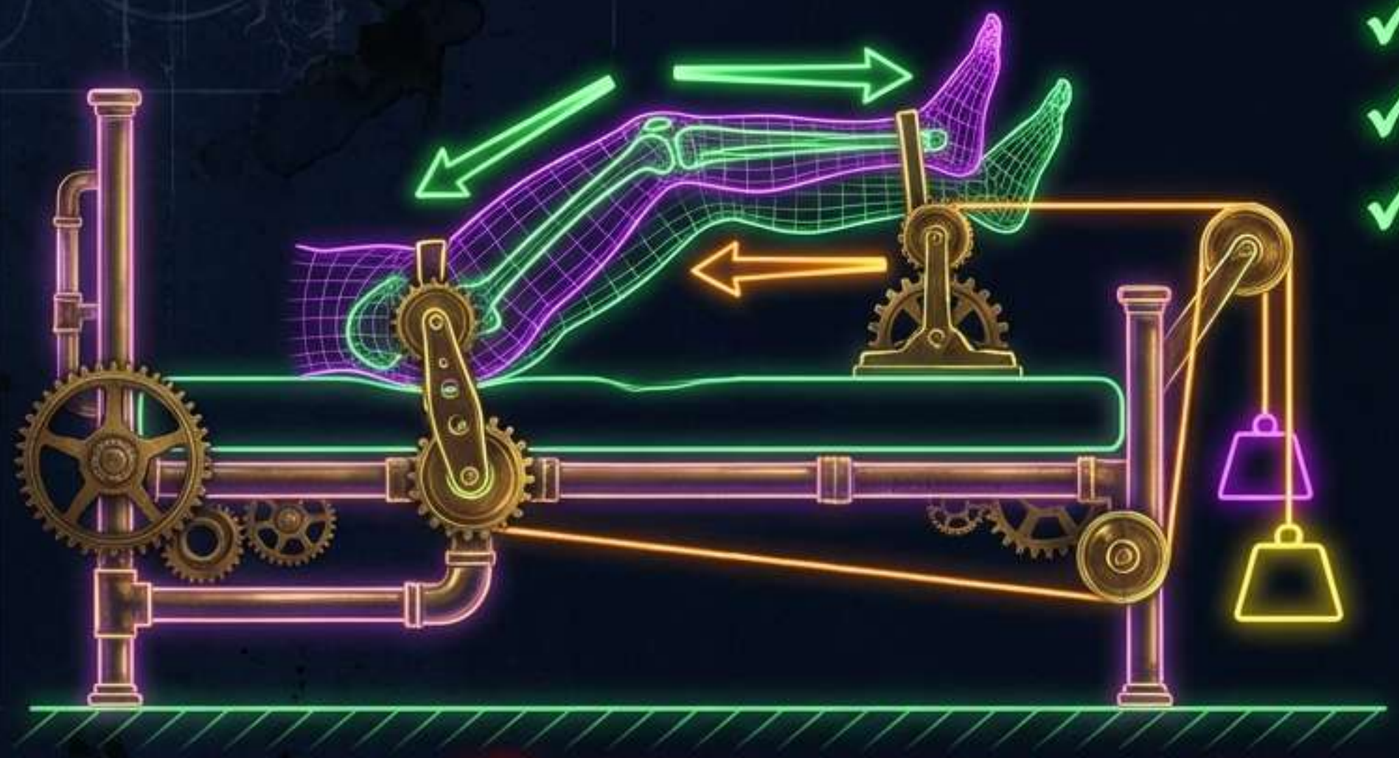
Realignment: Traction Physics



Share Tech Mono

- ✓ Tension: Continuous.
- ✓ Weights: Must Hang FREE.
- ✓ Knots: Secure.

⚠ CRITICAL: Weights must NOT touch floor.

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Component Resection: Phase I

Timeframe: 0 – 24 Hours.

ACTION: ELEVATE STUMP.

Objective: Reduce Swelling / Edema.

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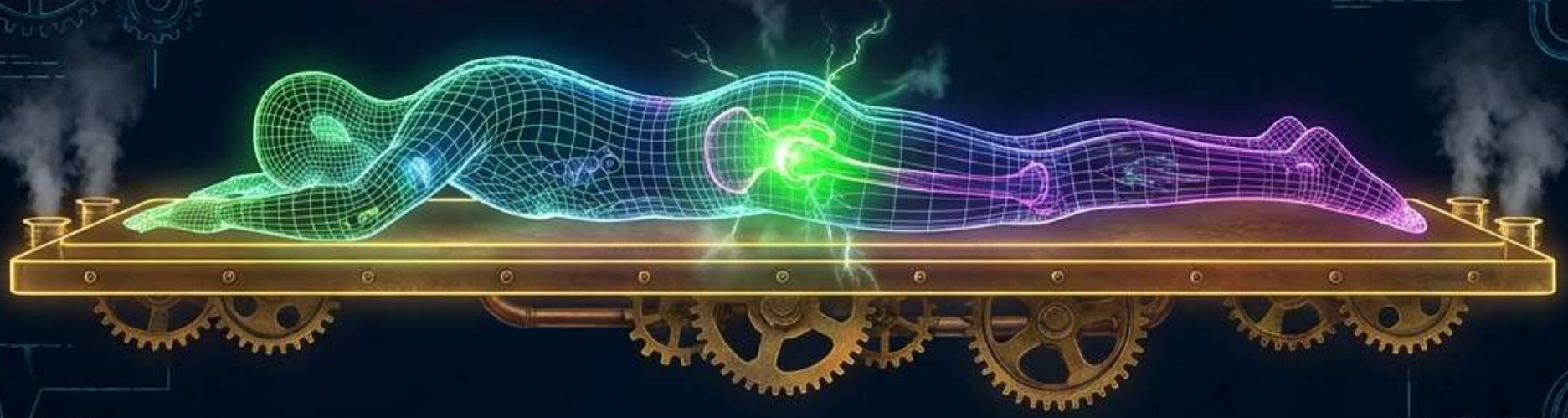
Component Resection: Phase II

Timeframe: > 24 Hours.

Risk: Hip Flexion Contracture.



ACTION: LIE PRONE daily (Extension).

⚠ DO NOT ELEVATE after 24 hours.

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Sensory Glitch: Phantom Pain

Status: EXPECTED / NORMAL.

Action: Treat pain with medication.

⚠ DO NOT treat as psychosis.

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Degradation Analysis: Arthritis

OSTEOARTHRITIS
(Mechanical Wear)

RHEUMATOID ARTHRITIS
(Systemic Fire)

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Osteoarthritis (OA)

Pathology: Wear and Tear / Unilateral.

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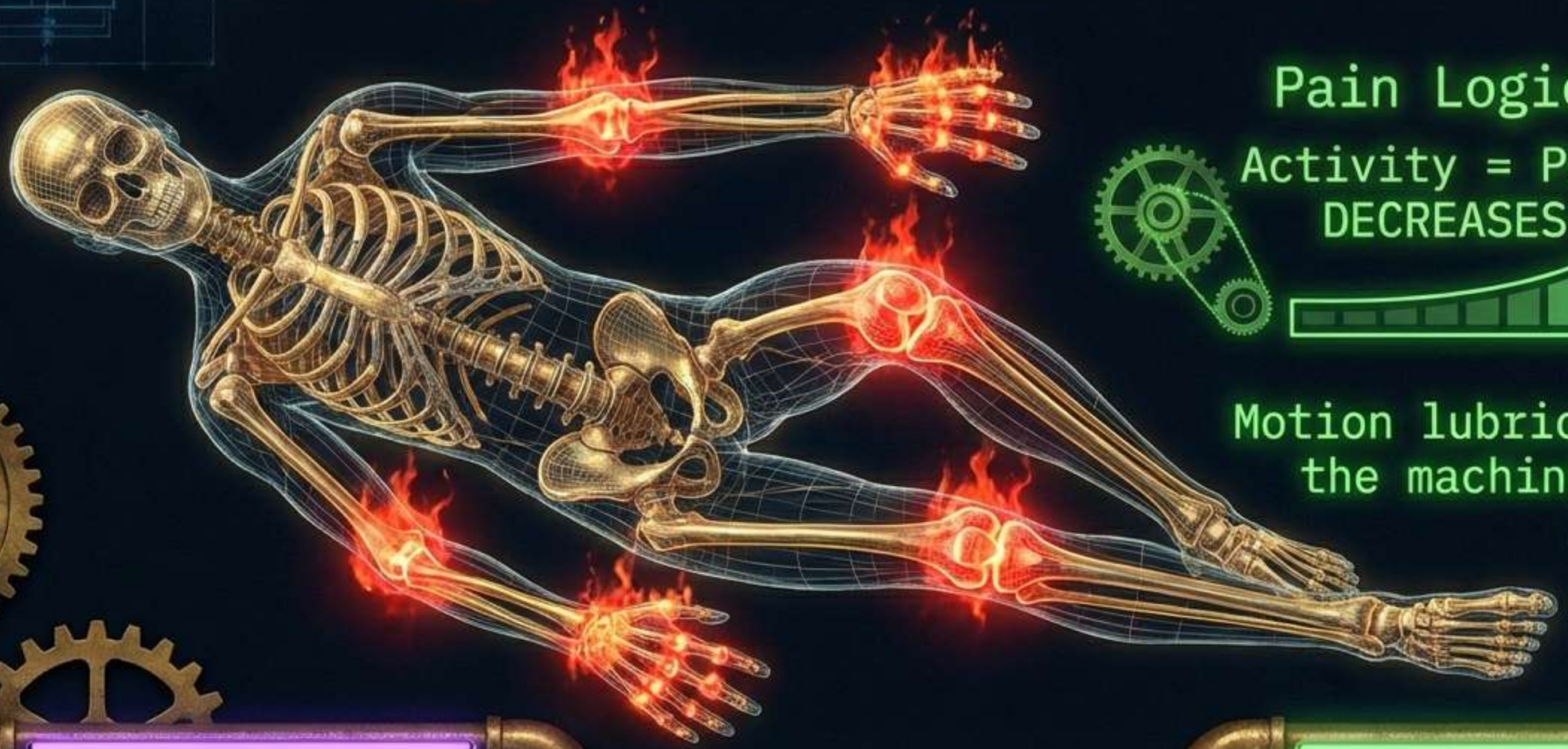
Rheumatoid Arthritis (RA)

Pathology: Systemic Autoimmune / Bilateral.



Pain Logic:

Activity = Pain DECREASES

Motion lubricates the machine.

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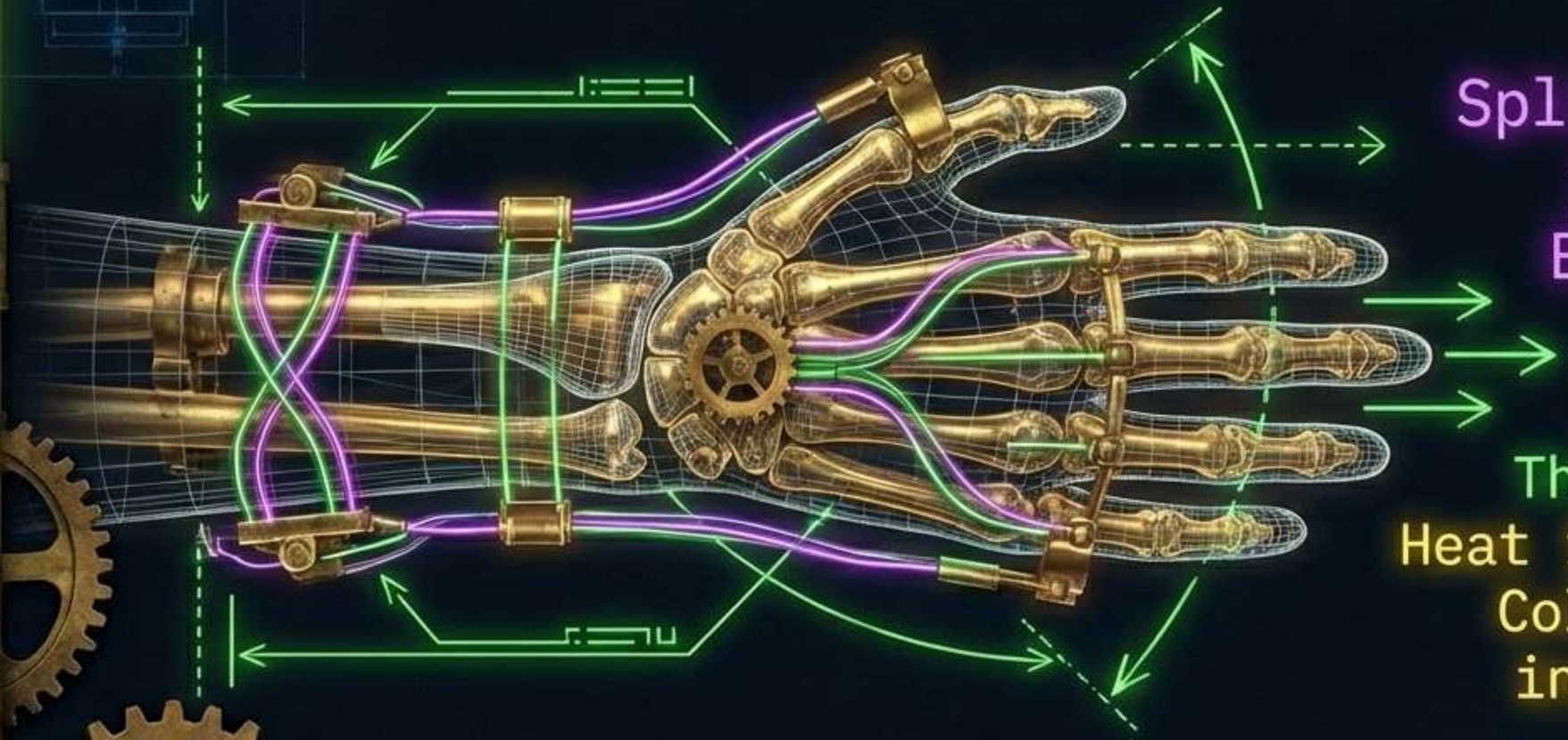
RA Intervention Protocols

Goal: Prevent Flexion Contracture.



Splint involved joints in EXTENSION.

Therapeutics:
Heat for stiffness / Cold for acute inflammation.

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Chassis Repair: Laminectomy

Handling: LOG ROLL only.

Mobilization Rules:

- NO Sitting > 30 Minutes.
- NO Dangling legs.
- Walk, Stand, or Lie Flat.

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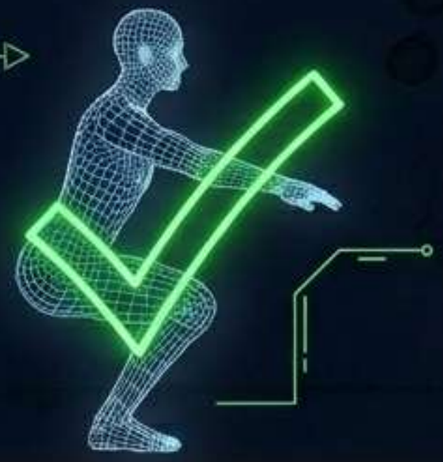
POST-OP LOAD BEARING LIMITS



TEMPORARY (6 Weeks)

- No driving.
- No lifting > 5 lbs.

PERMANENT (FOREVER)

- NEVER lift bending at waist (Use Knees).
- Cervical Laminectomy: NEVER lift above head.

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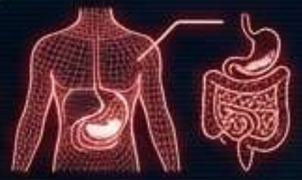
Critical System Alerts (Recap)

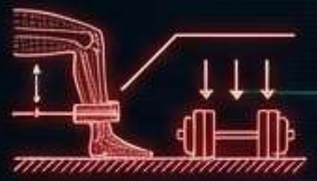
1. Crutches: Resting on Axilla (Nerve Damage).

2. Casts: Nausea/Vomiting (Cast Syndrome).



3. Amputation: Hip Flexion (Failure to lie Prone).

4. Traction: Weights on floor.



5. Spine: Sitting > 30 mins.

END TRANSMISSION.

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